NASA SPACE APPS CHALLENGE 2026

COSMIC TIME MACHINE

Our ambition, MVP scope, and the targets we will deliver together.

THE ONE-SENTENCE VISION

> **Make the universe easier to explore by letting anyone select a real region of the sky, see astronomical imagery, discover what observations exist, and understand what the data can tell us.**

1	1	1	3
real sky atlas	archive integration	complete demo journey	core user actions

Our guiding principle

We are not trying to build a complete astronomy platform in one hackathon. We are building a focused, credible experience that connects exploration to real data—and clearly distinguishes verified science from illustration.

Prepared for team alignment • Working scope as of 26 October 2026

1. Our ambition

Cosmic Time Machine should turn a complicated astronomy-data workflow into an intuitive journey. A user should not need to know archive tools, FITS files, or scientific catalogues before taking their first step.

The experience we want to create

- **Explore:** Start with an interactive view of the sky and navigate to an interesting region.
- **Discover:** Select a location or search by object name / celestial coordinates.
- **Connect to real observations:** Find relevant public survey imagery and SPHEREx archive products where available.
- **Understand:** See observation details, wavelength context, and a plain-language explanation of what is known.
- **Share:** Export a region summary or image with source information so another person can retrace the result.

Who we are building for

AUDIENCE	NEED	WHAT SUCCESS LOOKS LIKE
Curious public	Explore space without specialist knowledge.	They can find a region and explain one thing they learned.
Students / educators	A visual way to connect concepts with evidence.	They can use a region as a short learning activity.
Astronomy enthusiasts	Reach survey imagery and observation metadata.	They can inspect data provenance and open the source archive.

What makes our project distinctive

A guided, visual journey from a sky location to actual public observations, with scientific context and honest provenance. The value is not just showing a pretty space image; it is helping people understand where it came from and what they can learn from it.

2. The MVP we will deliver

MVP definition: a user can choose a real sky region, view real survey imagery, query public SPHEREx data for relevant observations, inspect the result or a clear no-results state, and leave with a source-linked summary.

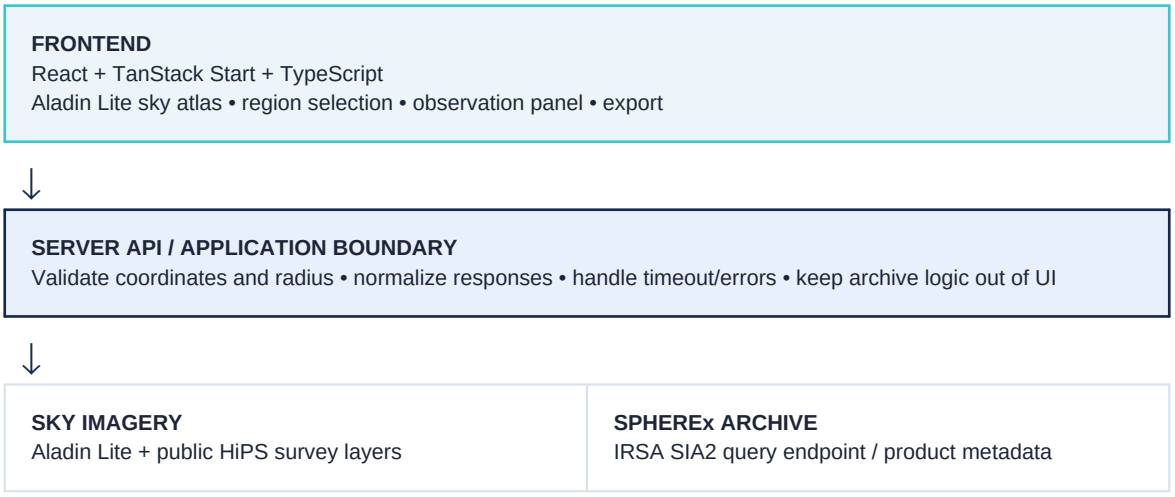
PRIORITY	CAPABILITY	MVP ACCEPTANCE CHECK
Must have	Interactive all-sky atlas	User can pan/zoom and select a region or enter coordinates.
Must have	Real survey imagery	At least one genuine survey layer is displayed with required attribution.
Must have	SPHEREx archive discovery	A region query returns real product metadata or an explicit no-results/error state.
Must have	Observation details	Show available product identifiers, observation metadata, source link, and coverage context.
Must have	Scientific integrity	Clearly label real data, illustrative UI, and any simulated/derived visualization.
Should have	Export / share	Download or copy a compact region summary with coordinates and source/provenance.
Should have	Learning context	Plain-language explanation of wavelength/survey and what the selected view represents.

Explicitly out of scope for the first MVP

- A full scientific analysis suite or professional-grade FITS processing pipeline.
- A guaranteed time-lapse of the same sky region across epochs; this requires compatible observations and careful validation.
- User accounts, social features, collaborative workspaces, or a production database unless a real MVP blocker emerges.
- AI-generated scientific conclusions that are not grounded in retrieved data and traceable sources.
- Claiming a region has SPHEREx coverage unless the archive response confirms it.

3. System architecture

Keep the first release small: a modular web application with a clear server-side boundary for archive queries. Avoid microservices, a database, or a separate processing service until the product demonstrates that it needs them.



Proposed internal API contract

ENDPOINT	PURPOSE
GET /api/observations?sra=&sdec;=&radius;=	Find matching SPHEREx archive products for a selected sky region.
GET /api/observations/:id	Return normalized metadata for a selected product, where supported.
GET /api/surveys	Return the supported survey layers and attribution information.

Architecture decisions

- **No database for MVP:** use live public archive results; add caching only if latency or rate limits make it necessary.
- **Use server-side proxying:** normalize the archive response and make timeouts/errors predictable for the frontend.
- **Use real survey data:** the atlas may use public HiPS imagery; SPHEREx product availability is checked separately.
- **Future FITS processing:** introduce a Python/Astropy worker only if creating custom composites or spectral plots becomes a must-have.

4. Delivery plan and success measures

The challenge weekend is 14–15 November 2026. As of 26 October, the team has roughly 19 days to validate the core path, finish the experience, and rehearse a reliable demonstration. The sequence below is a proposed plan, not a claim that the work is already complete.

WHEN	MILESTONE	DONE WHEN
Oct 26–28	Real sky atlas	Atlas loads, attribution is visible, user can select a region.
Oct 29–Nov 2	Archive query	Coordinate query returns parsed SPHEREx products or honest no-results.
Nov 3–7	Observation explorer	User can inspect metadata, understand the result, and open source records.
Nov 8–11	Polish & validate	Responsive layout, loading/error states, provenance, export, and test pass.
Nov 12–13	Demo rehearsal	A short, repeatable demo works on a clean session with fallback screenshots/data states.
Nov 14–15	Challenge weekend	Submit the project and present the real user journey clearly.

Definition of success

TARGET	MEASURE	WHY IT MATTERS
One end-to-end journey	From sky selection to archive result and source link.	Proves the core product promise.
Real, traceable data	At least one genuine sky survey layer; real archive results where available.	Avoids a demo that is only visual simulation.
Reliable fallback	No-data, timeout, and service-error states are clear and useful.	The experience remains credible when data is absent.
Explainable output	Coordinates, source, observation/product metadata, and data status are visible.	Users can trust and revisit the result.
Demo-ready	Team can complete a 2–3 minute walkthrough without relying on hidden steps.	Makes the ambition understandable to judges.

The most important target

> **By demo day, a user should be able to select a real location in the sky, see genuine survey imagery, check whether SPHEREx observations are available, understand the result, and follow a source link. If data is unavailable, the product must explain that honestly rather than inventing a match.**

5. Risks, guardrails, and team decisions

RISK	GUARDRAIL / RESPONSE
No SPHEREx product matches the selected region.	Treat no-results as a valid result; demo a verified region with coverage if available.
Archive response is slow or malformed.	Add timeout, loading state, normalized parsing, and a retry option; never show fabricated metadata.
Images from different surveys are compared incorrectly.	Show survey, wavelength, and coverage labels; avoid implying a scientific time-lapse without validation.
Scope grows into a full research platform.	Protect the MVP checklist; defer accounts, advanced processing, and AI analysis.
Illustrative UI is mistaken for science data.	Label mock visuals as illustrative; label live survey imagery and archive metadata separately.

Decisions the team should agree on now

- The MVP success definition and the must-have list on page 3.
- Who owns the sky atlas integration, archive/API integration, UI/UX, and testing/demo narrative.
- One verified example region to use for the main demo, plus a fallback if live archive access fails.
- A rule that every scientific-looking output carries a clear source and data-status label.
- A short check-in cadence and a code freeze before the final demo rehearsal.

Suggested 2–3 minute demo narrative

1	Choose a region	Start from the real sky atlas and select a point of interest.
2	Explore imagery	Show the actual survey layer and explain what it represents.
3	Check SPHEREx	Query the archive and inspect a product or a transparent no-results response.
4	Take learning away	Read the region summary, see provenance, and export/share the result.

Closing statement

Our ambition is big: make the sky feel explorable and science feel approachable. Our target is deliberately focused: deliver one trustworthy end-to-end experience that proves the idea with real data. Build the smallest version that earns trust, then expand.

Reference points for implementation: Aladin Lite documentation (aladin.cds.unistra.fr/AladinLite/doc/); SPHEREx data access documentation (caltech-ipac.github.io/spherex-archive-documentation/); IRSA SPHEREx archive (irsa.ipac.caltech.edu/Missions/spherex.html). Confirm current API parameters, survey identifiers, and product availability during implementation.